

GLASGOW COLLEGE UESTC

Exam paper

Digital Signal Processing (UESTC4005)

Date: 22nd June 2022

Time:09:30-11:30

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

**Make sure that your University of Glasgow and UESTC Student Identification
Numbers are on all answer sheets.**

**An electronic calculator may be used provided that it does not allow text storage
or display, or graphical display.**

DATA/FORMULAE SHEET IS PROVIDED AT THE FRONT OF PAPER

Table 1 DFT Theorems

Theorem	Length- N Sequence	N -point DFT
	$g[n]$ $h[n]$	$G[k]$ $H[k]$
Linearity	$\alpha g[n] + \beta h[n]$	$\alpha G[k] + \beta H[k]$
Circular time-shifting	$g[(n - n_o)_N]$	$W_N^{kn_o} G[k]$
Circular frequency-shifting	$W_N^{-k_o n} g[n]$	$G[(k - k_o)_N]$
Duality	$G[n]$	$N g[(-k)_N]$
N -point circular convolution	$\sum_{m=0}^{N-1} g[m] h[(n - m)_N]$	$G[k] H[k]$
Modulation	$g[n] h[n]$	$\frac{1}{N} \sum_{m=0}^{N-1} G[m] H[(k - m)_N]$
Parseval's theorem	$\sum_{n=0}^{N-1} g[n] ^2 = \frac{1}{N} \sum_{k=0}^{N-1} G[k] ^2$	

Table 2 The parameters of fix windows

Type of window ($w[n]$, $-M \leq n \leq M$)	Minimum stopband attenuation(α_s)	Transition bandwidth ($\Delta\omega$)
Rectangular	20.9dB	$0.92\pi/M$
Hann	43.9dB	$3.11\pi/M$
Hamming	54.5dB	$3.32\pi/M$
Blackman	75.3dB	$5.56\pi/M$

Some Fixed window functions

$$\text{Hann: } w[n] = \frac{1}{2} \left[1 + \cos \left(\frac{\pi n}{M} \right) \right], \quad -M \leq n \leq M,$$

$$\text{Hamming: } w[n] = 0.54 + 0.46 \cos \left(\frac{\pi n}{M} \right), \quad -M \leq n \leq M.$$

$$\text{Blackman: } w[n] = 0.42 + 0.5 \cos \left(\frac{\pi n}{M} \right) + 0.08 \cos \left(\frac{2\pi n}{M} \right), \quad -M \leq n \leq M.$$

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- Q1 A linear time-invariant system is described by the difference equation:
 $y[n] = \sum_{k=0}^7 a_k x[n-k]$, where $a_k = \{-3, 5, 1, 6, -6, -1, -5, 3\}$, $k = 0, 1, 2, 3, 4, 5, 6, 7$
- Determine the impulse response $h[n]$ of the system. [4]
 - Is the system an FIR or IIR? Why? [5]
 - Is it a linear phase system? If it is, Which type is this linear phase system? Why? [5]
 - If $H(z)$ is the transfer function of the system, can we definitely say that $H(z)$ has zeros at $z=1$ and $z=-1$? Please give detailed reasons. [6]
 - If $H(e^{j\omega})$ is the frequency response of the system, calculate $\int_{-\pi}^{\pi} H(e^{j\omega}) d\omega$. [5]
- Q2 Consider a finite-length sequence $x[n] = \{1, 0.5, 0.75, 0.25, 1, -1, -0.125, 0.25\}$ $0 \leq n \leq 7$. The 8-point DFT of $x[n]$ is denoted as $X[k]$.
- Given a sequence $y[n]$ with 8-point DFT being $Y[k] = W_4^k X[k]$, Determine and sketch $y[n]$. [6]
 - Draw a diagram showing how $X[k]$ may be calculated using a radix-2 decimation in Time Fast Fourier Transformation algorithm (DIT FFT) [6]
 - If it is desired to evaluate $X(e^{j\omega})|_{\omega=0.4\pi}$ and $X(e^{j\omega})|_{\omega=0.5\pi}$ by the same N-point DFT directly, determine the smallest value of N. [5]
 - Given a sequence $z[n]$ described by $z[n] = \begin{cases} x[n/2], & n = \text{even} \\ 0, & n = \text{odd} \end{cases}$. The 16-point DFT of $z[n]$ is $Z[k]$. Determine the expression of $Z[k]$ in terms of $X[k]$. [8]
- Q3 Consider a system shown in Fig3.1, the frequency response of $H(e^{j\omega})$ of a digital system and the input signal spectrum $X_c(j\Omega)$ are indicated in Fig 3.2, respectively:

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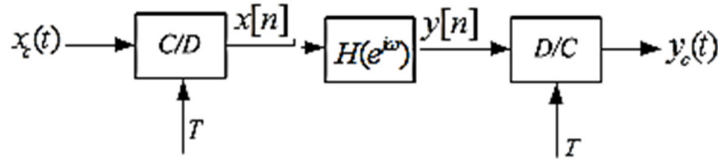


Fig 3.1

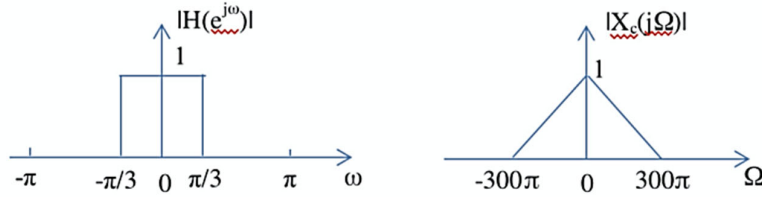


Fig 3.2

- (a) If we want to avoid frequency spectrum aliasing in the C/D converter, what is the maximum value of T should be chosen? [4]

- (b) If $T = \frac{1}{500} s$, $H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \frac{\pi}{3} \\ 0, & |\omega| > \frac{\pi}{3} \end{cases}$, sketch the spectrum of $x[n]$, $y[n]$ and $y_c(t)$ (Note: Pay attention to the amplitude of each spectrum). [12]

- (c) If the value of T in C/D converter is still equal to $\frac{1}{500} s$, but the value of T in D/C converter is changed to be $\frac{1}{800} s$, sketch the spectrum of $x[n]$, $y[n]$ and $y_c(t)$ (Note: Pay attention to the amplitude of each spectrum) [9]

Q4 Consider a signal $x(t) = A\cos(2\pi f_1 t) + B\cos(2\pi f_2 t)$, $f_1 = 5\text{kHz}$, $f_2 = 8\text{kHz}$, Suppose $x(t)$ is sampled with $f_T = 20\text{kHz}$ to produce the discrete sequence $x[n]$.

- (a) Write down an expression for $x[n]$. [4]
- (b) If $A = B = 1$, based on the Window Fourier Series Method to design a linear phase FIR Low-pass filter (choosing a fixed window) with the least order to block the component f_2 by at least 45dB: [15]

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- (i) Suppose the pass-band edge frequency $f_p=6000\text{Hz}$, stop-band edge frequency $f_s=7500\text{Hz}$, determine the cutoff frequency ω_c of the ideal discrete-time LPF. [4]
- (ii) Give the expression of the $h_{id}[n]$ of the ideal linear-phase LPF. [4]
- (iii) Choose the type and length of window to meet the specifications and costs minimum resources (i.e. the smallest number of multipliers, adders and delayers). [4]
- (iv) Write down the impulse response $h[n]$ of the designed linear phase FIR LPF. (Note: It should be causal.) [3]
- (c) Sketch a realization with the fewest multipliers. [6]